在线代理模型研究：工作逻辑、结果与下一步

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目标与实验逻辑

目标是在逐轮获取 solver 真值的在线过程中，比较代理模型帮助寻找低 R 权重 W 的能力。五个模型从相同的 2048 条初始数据出发，在每轮共同的候选池上按预测均值最低选点；solver 返回真实 R 后，只反馈给该模型自身并重训。这样保持候选选项一致、模型轨迹独立。在线寻优和预测误差分别评价，测试集题目与标签应在策略冻结前封存。

候选池经历三版：V1 每池 50 个全局 Dirichlet 点 + 50 个低值训练点附近扰动；V2 每池 60 个全局点 + 240 个围绕初始前 25 个低值点的多尺度扰动；V3 按润泽提出的离线漫游思路，围绕最佳初始权重 $p01969$，沿离线筛选方向及更广方向，以步长 0.01–0.05 生成每池 300 点。当前统一选点规则为最小预测均值；离线方向只是候选生成先验，真实优劣由 solver 验证。

已完成实验和模型表现对比

V1 pilot 中 5 模型各运行 5 轮，仅得到 11 个不同选点，最佳值只从 115.765 微降至 115.759；100 点开发测试没有覆盖 $R \leq 115.68$。近优压力集 50 点中只有 7 点落入 $R \leq 120.765$，local-GP 在这 7 点的 MAE 最低（1.192），但标签已揭晓。

模型	V2 最佳 R 均值 ±SD（命中/3）	V3 最佳 R 均值 ±SD（命中/3）	拟合秒/轮 V2 / V3
local-GP	113.820 ± 0.534（3/3）	114.238 ± 0.464（3/3）	56.0 / 48.0
HistGB	113.950 ± 1.577（2/3）	113.915 ± 0.090（3/3）	6.3 / 4.04
XGBoost	113.959 ± 0.785（3/3）	114.290 ± 0.827（3/3）	3.0 / 2.09
boosted-consensus	114.287 ± 1.057（3/3）	114.434 ± 0.511（3/3）	9.3 / 6.11
ridge2	115.765 ± 0.000（0/3）	114.163 ± 0.350（3/3）	0.5 / 0.29

解读：V2 多尺度池中 local-GP 均值最好、XGBoost 拟合快且表现接近；V3 漫游池中 HistGB 均值最低且最稳定，XGBoost 得到跨两版最低单点 $R = 113.444$。ridge2 从 V2 的 0/3 命中变为 V3 的 3/3。每版内五模型共享相同候选池；V2 与 V3 改变了候选生成方式，跨版差异不能解释为纯模型因果差异。

数据资产与验证边界

The deduplicated solver database now contains 1,370 unique non-training weights, with no overlap against the original 2,048 rows. It includes 152 unique V3.1 model-selected points and 799 new points from the global audit; all other V1/V2/V3 and pilot sources remain indexed by provenance. The audit labels were excluded from training. Earlier pilot and stress labels are development data, not a blind test. V3.1 had no test set. Online solver time was 36.7 min; separate global audit time was 2.62 h.

V3.1 results and candidate evidence

V3.1: 8 groups x 10 rounds, 5 models, strict 2048-row cold start; each model receives only its own chosen solver feedback. Same pool per round; first-place credit is fractionally split for ties. No test set; no full-pool MAE or Spearman.

Model	Mean group minimum	Lowest R	Wins/80	Fit seconds
GP-ARD	**112.678**	112.161	**33**	10,146
HistGB	112.724	**111.704**	16	438
Boosted-consensus	112.976	112.239	14	648
XGBoost	113.044	112.344	14	211
Local-GP	114.196	112.897	3	3,711

GP-ARD led the average of 8 group minima and same-round wins, with the longest fit time and some boundary convergence warnings. HistGB found the lowest selected point (R=111.704) and fit in 438 s. V3.1 costs: 4.21 h total model fitting, 36.7 min online solver, and 2.62 h global audit. Candidate pool: 10 global Dirichlet + 290 local points; local anchors R=113.444/112.907/113.951; steps 0.01–0.10, denser at 0.01–0.06. Shortlist and broad each supplied 48.33% of local appearances; selection shares were 76.75% and 23%. All 800 global points were solved separately: minimum 117.262, none below 115, only one selected online. V3 low R=113.444 came from a broad screened direction. V3.1 low R=111.704 came from a broad feasible direction at step 0.09 around the R=112.907 anchor; shortlist also yielded R=112.239. Selected-point outcomes are biased by model choice and do not prove source-wide superiority or an optimal step. Next: balance solver coverage across source, anchor, and step, then compare selectors.

详细设计、逐轮日志、报告、候选池和原始调用记录见 `analysis/online_model_study_v1/`；唯一数据表位于 `solver_database/`。本报告数值来自已归档开发实验，结论限于所测试的候选生成方式与三个随机种子。